

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2024
<i>Expected Completion Date:</i>	May 2025
<i>Project Title:</i>	LLM-based spreadsheet prompting tool
<i>Company:</i>	Please provide company name
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Machine Learning, Large Language Models
Project Background Information	
Excel is used ubiquitously by individuals and organisations to handle small and medium-sized datasets. It is an extremely versatile tool, well-suited for storing and processing structured data in limited quantities, particularly if the data structure and/or tasks change frequently. However, its flexibility can be both its strength and its weakness. It is relatively easy to introduce errors while processing data, particularly if the data is being frequently updated or when it is incomplete or inconsistent. Furthermore, Excel formulas can become complex if the data comes from several sources or the logic is non-trivial. Consider the following example. The user is provided with: - A spreadsheet containing the list of academics in the department; - A list of academic roles allocated to each academic, one of which is "personal tutor"; - A list of form responses in which academics provided their availability for tutor meetings; - A list of students and their assigned tutor; - A table assigning a time slot and location to each tutor meeting.	

¹ An overview of different research areas in the school of computer science can be found [here](#).

A user may want to create a new Excel spreadsheet that lists each tutor's name, the time and location of their tutor meeting (or a note if they are unavailable), along with the list of their tutees.

They may also want to create another Excel spreadsheet that calculates the number of meetings in each room for each time slot, to ensure there are no clashes.

There may be imperfections in the data. Not every academic completed the form, and the names of academics are written in slightly different formats across the source spreadsheets. There are inconsistencies in academic lists as well as spelling discrepancies in room names, e.g., "C80" or "C 80".

Requirements

The group must provide:

You will develop a tool to assist users in processing Excel data. Your tool will take one or more Excel spreadsheets as the input, along with a user prompt in a natural language.

An example of the prompt:

"There is a list of academics in File 1. There is also a list of roles assigned to each academic in File 2. Choose the academics who are assigned the role 'personal tutor'; these are personal tutors. Produce a table of all personal tutors. It should have the columns Tutor, Meeting location, Meeting time, and List of tutees. Tutor is the name of the personal tutor, Meeting location and Meeting time are taken from File 2, and the List of tutees is the list of students assigned to this academic as their tutor in File 3. If a personal tutor is not available for any time slot (see columns Available at 1 pm, Available at 2 pm, etc. in File 4), put 'Unavailable' in the Meeting time column and leave Meeting location empty."

Your tool will use a large language model (LLM) to convert the prompt into a sequence of atomic commands. It will then execute this sequence to produce a new spreadsheet. For example, the LLM may convert the above prompt into commands such as "Join table in File 1 with table in File 2 based on columns 'Academic' and 'Name' and save it as Table 5. Remove rows in Table 5 if Column 'Personal tutor' has value 'N'. ..."

Your tool should implement the following functionality:

- Accept multiple Excel files as input.
- Accept a user prompt in a natural language.
- Use an LLM to convert the user prompt into a sequence of atomic commands.
- Execute the atomic commands.
- Save the output in a new Excel spreadsheet.
- Provide feedback on the clarity of the user prompt.
- Flag issues with the data.
- Handle imperfect data such as minor inconsistencies in spelling.

Optional functionality:

- Support prompts that ask to produce visualisations.

- Support prompts that ask for conditional formatting in the output.
- As an alternative to executing the atomic commands within your system, produce an Excel file that uses Excel formulas to execute those commands.

Key expectations:

- The tool should be user-friendly and require no prior experience.
- The tool should help users avoid errors.
- The tool should handle imperfect data (e.g., slight spelling or formatting discrepancies) where native Excel commands may fail.
- The tool should tolerate minor modifications to the input data format (e.g., reshuffled columns) without requiring updates to the user prompt.
- The tool should not share sensitive data with third-party services.

Skills

Highly desirable:

Experience of GUI development
Experience of manipulating data
Experience of using large language models

Desirable:

Submission / Award information

Date of Submission:

September 2025

Date of Pitch:

October 2025

Notification of award:

October 2025

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.